

# Jatin Chowdhury

## Audio Signal Processing Engineer

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## Education

### Massachusetts Institute of Technology

Cambridge, MA

Ph.D in Electrical Engineering and Computer Science

Sept. 2025 - PRESENT

- Relevant research: real-time circuit simulation, audio software development, signal processing.
- Relevant coursework: numerical algorithms, TinyML.

### Stanford University, Center for Computer Research in Music and Acoustics

Palo Alto, CA

M.A. in Music, Science, and Technology

Sept. 2018 - June 2020

- Relevant coursework: physical audio signal processing, time-frequency audio signal processing, machine learning, digital instrument design, spatial audio, music recording.
- Relevant research: physical modelling, nonlinear filter design, modal modelling, machine learning.
- Denning Family Fellowship for the 2018-2019 academic year.

### University of Southern California

Los Angeles, CA

B.S. in Electrical Engineering

Aug. 2014 - May 2018

- Completed Minors in Physics and Music Recording.
- Relevant coursework: signal processing, circuit design, digital logic, electromagnetics, software design.
- USC Presidential Scholarship for outstanding academic achievement.
- USC Renaissance Scholar certificate for excelling academically while pursuing separate fields of study.

## Experience

### Chowdhury DSP

Renton, WA

Audio Signal Processing Engineer

Oct. 2021 - PRESENT

- Developed signal processing algorithms for real-time audio processing and synthesis.
- DSP algorithms include: virtual analog modelling, machine learning, filter design, and frequency-domain processing.
- Developed audio plugins including DSP, UI implementation, preset management, copy protection, installers, etc.
- Rendered services for clients including: BABY Audio, Schwabe Digital, Chandler Limited, Bogren Digital, Little Sir Rekko, VCV, KIT Plugins, Sabol Hill, Submission Audio, ADPTR Audio, and more.
- Hired, mentored, and trained interns in audio signal processing and software development.

### Tesla Motors

Palo Alto, CA

Audio Test Engineer

Aug. 2020 - Oct. 2021

- Developed audio system signal flow layouts for vehicles using AudioWeaver.
- Contributed to audio system testing and tuning including telephony, equalization, and spatialization.
- Developed end-of-line tests to ensure vehicle audio system quality using Python and C++.
- Contributed to the signal processing and tuning for the vehicle pedestrian warning speaker.
- Contributed to testing and validation software for audio system firmware.

### Persp3ctive VR

Los Angeles, CA

Software/DSP Engineer

June 2019 - Mar. 2020

- Developed and implemented audio effects for use in a VR audio production environment.
- Implemented the audio effects as an audio plugin with a designed GUI.
- Built DSP framework for integrating effects made by other engineers into the plugin.
- Implemented OSC communication for sharing parameters and metering data with VR headset.

### Audioworks Technologies

Toronto, ON

Software Engineer

July 2018 - Dec. 2018

- Member of the SoundsUnite development team: building a digital audioworkstation (DAW) using JUICE/C++.
- Developed DSP features for the application including panning algorithms, level detection, and "smart" track exporting.
- Developed UX features for the application including meters, file browsers, and tempo controls.
- Integrating the application file management system with the SoundsUnite web store.

### McGill Space Institute

Montreal, QC

Research Trainee

May 2017 - Aug. 2017

- Member of the CHIME/FRB Working Group: building a software pipeline to detect Fast Radio Bursts (FRBs).
- Developed a "Flux Estimator" module for the pipeline to estimate the intrinsic brightness of astrophysical events.
- Contributed to science and unit testing frameworks for the CHIME/FRB software pipeline.
- Presented work for the CHIME/FRB Working Group and CHIME/FRB Pipeline Gamma Release.

# Skills

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|------------------------------|------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | C/C++, Python, Bash, Jai, MATLAB, Faust, LaTeX                                     |
| <b>Signal Processing</b>     | Virtual analog modelling, filter design, machine learning, time-frequency analysis |
| <b>Frameworks</b>            | JUCE, CLAP, TensorFlow, PyTorch, Arduino                                           |
| <b>Tools</b>                 | CMake, Git, Linux CLI, Visual Studio, Xcode, GitHub Actions, Inno Setup            |

# Publications

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| <b>Performance-Oriented Wave Digital Circuit Emulation</b><br>Proc. of the 29th International Conference on Digital Audio Effects<br>• Co-authored with Mark Rau.<br>• Available on the DAFx Archives.                                                                                   | Sept. 2026<br>Cambridge, USA       |
| <b>Inference-Time Structured Pruning for Real-Time Neural Network Audio Effects</b><br>Proc. of the 28th International Conference on Digital Audio Effects<br>• Co-authored with Christopher Johann Clarke.<br>• Available on the DAFx Archives.                                         | Sept. 2025<br>Ancona, IT           |
| <b>Sample Rate Independent Recurrent Neural Networks for Audio Effects Processing</b><br>Proc. of the 27th International Conference on Digital Audio Effects<br>• Co-authored with Alistair Carson, Alec, Wright, Vesa Välimäki, and Stefan Bilbao.<br>• Available on the DAFx Archives. | Sept. 2024<br>Guildford, UK        |
| <b>Computationally Efficient Physics Approximating Neural Networks for Highly Nonlinear Maps</b><br>Proc. of the Conference on Research in Adaptive and Convergent System<br>• Co-authored with Christopher Johann Clarke and others.<br>• Available on the ACM Digital Library.         | Oct. 2022<br>New York, USA         |
| <b>chowdsp_wdf: An Advanced C++ Library for Wave Digital Circuit Modelling</b><br>ArXiv E-Prints: Audio and Speech Processing<br>• Presents a wave digital circuit modelling library with fixed-topology optimizations and other advancements.<br>• Available on the ArXiv.              | Oct. 2022<br>Online                |
| <b>Emulating Diode Circuits with Differentiable Wave Digital Filters</b><br>19th Sound and Music Computing Conference<br>• Co-authored with Christopher Johann Clarke.<br>• Available on Zenodo.                                                                                         | June 2022<br>Saint Etienne, France |
| <b>A Wave Digital Filter Modeling Library for the Faust Programming Language</b><br>Proc. of the 18th Sound and Music Computing Conference<br>• Co-authored with Dirk Roosenburg, Eli Stine, and Romain Michon.<br>• Available on GitHub.                                                | June 2021<br>Torino, IT            |
| <b>RTNeural: Fast Neural Inferencing for Real-Time Systems</b><br>ArXiv E-Prints: Audio and Speech Processing<br>• Presents a neural network inferencing library for real-time systems.<br>• Available on the ArXiv.                                                                     | June 2021<br>Online                |
| <b>Water Bottle Synthesis with Modal Signal Processing</b><br>Proc. of the 23rd International Conference on Digital Audio Effects<br>• Presented at the DAFx-2020 conference.<br>• Available on the DAFx Archives.                                                                       | Sept. 2020<br>Vienna, Austria      |
| <b>Stable Structures for Nonlinear Biquad Filters</b><br>Proc. of the 23rd International Conference on Digital Audio Effects<br>• Presented at the DAFx-2020 conference.<br>• Available on the DAFx Archives.                                                                            | Sept. 2020<br>Vienna, Austria      |
| <b>Real-Time Physical Modelling for Analog Tape Machines</b><br>Proc. of the 22nd International Conference on Digital Audio Effects<br>• Presented at the DAFx-2019 conference.<br>• Available on the DAFx Archives.                                                                     | Sept. 2019<br>Birmingham, UK       |

- Co-authored with the CHIME/FRB Collaboration
- Available on the ArXiv.

## Professional & Academic Service

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|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Peer Reviewer</b> | Provided peer review for papers submitted to conferences and journals, including DAFx, Audio Mostly, and the EURASIP Journal on Audio, Speech, and Music Processing. |
| <b>Organizer</b>     | Served on the local organizing committee for the 29th International Conference on Digital Audio Effects (DAFx26), in Cambridge, MA.                                  |
| <b>Mentor</b>        | Participated in the Audio Developer Conference Mentorship Program in 2025 and 2026.                                                                                  |

## Presentations

### Wait-Free Programming From Scratch

June 2026

Northwest C++ User Group Meetup

Bellevue, WA

- Discussed the challenges of designing wait-free algorithms for audio and other real-time software applications.
- Discussed the trade-offs of well-known solutions including spin mutexes and lock-free queues.
- Presented the development of a “from scratch” wait-free solution to a specific problem.

### AI Inference for Real-Time Audio Applications

February 2026

Boston AI Music Meetup

Cambridge, MA

- Discussed the challenges of developing real-time systems in the context of music software.
- Discussed the challenges of performing neural network inference within a real-time system.
- Presented RTNeural as an inference engine capable of performing neural network inference in real-time.

### Performance Engineering for Modern DSP

September 2025

Signal Processing Summit

Milpitas, CA

- Discussed performance engineering for signal processing engineers and software developers.
- Presented recommendations based on modern advancements in CPU and DSP hardware technology.
- Discussed ideas for teaching novice engineers about performance.

### Real-Time Audio Programming with Memory Arenas

October 2024

The Audio Programmer Monthly Meetup

Online

- Presented a brief overview of heap and stack memory allocation.
- Introduced the concept of the “arena allocator” and some of its properties.
- Discussed the use of arena allocators for real-time audio programming applications.

### Productive Programming with the JAI Programming Language

October 2024

Northwest C++ User Group Meetup

Redmond, WA, USA

- Presented a brief introduction to the JAI programming language.
- Discussed relevant JAI features, including compile-time meta-programming and the JAI `Context`.
- Discussed the use of JAI in audio programming applications.

### Real-Time Circuit Simulation with Wave Digital Filters In C++ (Poster)

September 2024

CppCon

Aurora, CO, USA

- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ compile-time meta-programming for implementing WDFs.
- Compared the runtime performance of various WDFs written in C++ and other languages.
- Presented a simple synthesizer made entirely with WDFs.

### A Comparison of Virtual Analog Modelling Techniques (Part 2)

November 2023

Audio Developer Conference

London, UK

- Co-presented with Christopher Johann Clarke.
- Presented a virtual analog model of the Tube Screamer distortion circuit using multiple modelling techniques.
- Discussed white-box methods: Modified Nodal Analysis, Nodal DK, and Wave Digital Filters with R-Type adaptors.
- Discussed black-box methods: Recurrent Neural Networks, conditional models, and transformers.
- Discussed grey-box methods: Symbolic Regression, Differentiable WDFs, and Physics-Approximating Neural Nets.

### Audio Buffers, Buffer Views, and Buffer Iterators

September 2023

The Audio Programmer Monthly Meetup

Online

- Reviewed data structure design in C++, focusing on owning and non-owning containers.
- Discussed the design of an “owning” audio buffer data structure, and a “non-owning” buffer view data structure.
- Discussed the implementation of custom C++ iterators for audio buffer data structures.

## Real-Time Circuit Simulation with Wave Digital Filters

July 2023

Northwest C++ Users Group Meetup

Redmond, WA

- Discussed the challenges of real-time audio programming.
- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ run-time polymorphism and compile-time meta-programming for implementing WDFs.
- Demonstrated example applications of WDF models in developing digital audio effects.

## Replacing the AudioProcessorValueTreeState

December 2022

The Audio Programmer Monthly Meetup

Online

- Discussed the historical and current methods of audio plugin state management with the JUCE framework.
- Proposed a new approach for managing plugin state with an emphasis on real-time safety.

## Physical Modelling for Analog Tape Emulation

December 2022

Acoustical Society of America Annual Meeting

Nashville, USA

- Discussed the development of a physical-modelling algorithm for real-time tape emulation.
- Discussed the physical basis for modelling magnetic hysteresis with nonlinear differential equations.
- Discussed implementation challenges for real-time tape emulation algorithms.

## Analog Modelling with Wave Digital Filters in C++

December 2021

The Audio Programmer Monthly Meetup

London, UK

- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Discussed two approaches for implementing WDFs in C++, with trade-offs between flexibility and performance.
- Demonstrated example applications of WDF models in developing digital audio effects.

## CI/CD for Audio Plugin Development

November 2021

Audio Developer Conference

London, UK

- Discussed the basic concepts and motivations for continuous integration and continuous deployment (CI/CD).
- Discussed the usefulness of CI/CD for developing audio plugins.
- Demonstrated several CI/CD workflows in the context of audio plugin development.

## Creating Audio Effect Plugins for the 21st Century

February 2021

SARC Virtual Events Series

Belfast, UK

- Presented at the Sonic Arts Research Centre (SARC) at Queen's University Belfast.
- Presented a paradigm for developing original and innovative digital audio software.
- Discussed techniques and philosophies for designing audio software as part of a creative process, including signal processing algorithms and user interfaces.

## A Comparison of Virtual Analog Modelling Techniques

November 2020

Audio Developer Conference

London, UK

- Presented a virtual analog model of the Klon Centaur guitar pedal developed using multiple modelling techniques, including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.
- An accompanying paper is published on the ArXiv.

## Complex Nonlinearities for Audio Signal Processing

May 2019

CCRMA DSP Seminar

Stanford, CA

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented a series of audio effects constructed from innovative uses of nonlinear signal processing techniques.
- Audio effects presented include harmonic exciters, hysteresis, nonlinear filters, subharmonics generators, and more.

## Exploring Real-Time DSP Systems for Mixing and Performance

Apr. 2019

CCRMA DSP Seminar

Stanford, CA

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented overview of previous research and projects to an audience of peers, professors, and the public.
- Discussed topics including real-time timbral conversion, methods of integrating "imperfection" into audio DSP systems, and designing user interfaces for DSP systems as an extension of the DSP systems themselves.

# Research

## Code Generation Workflows for Circuit Emulation

Cambridge, MA

Lead Researcher

June 2025 - PRESENT

- Developed code generation systems for generating optimized implementations of circuit models using either Wave Digital Filters or nodal analysis.
- Developed a textual representation of Wave Digital circuit topology.
- Developed analysis tools for "reducing" circuit model complexity.

## Real-Time Neural Network Inferencing

Stanford, CA

Lead Researcher

June 2020 - PRESENT

- Developed RTNeural, a C++ library for performing neural inferencing for real-time audio systems.
- Developed performance benchmarks to compare the library with LibTorch, ONNX Runtime, and TensorFlow Lite.
- Project documentation is available on GitHub.

## Differentiable Wave Digital Filters

Lead Researcher

July 2021 - Sept. 2022

- Demonstrated a process for constructing Wave Digital Filter (WDF) circuit models, in conjunction with neural network models of specific circuit elements.
- Developed a Python library for creating differentiable WDF models, allowing them to be trained via backpropagation.
- Project documentation is available on GitHub.

## Klon Centaur Emulation

Stanford, CA

Lead Researcher

Mar. 2020 - Aug. 2020

- Developed a digital emulation of the Klon Centaur guitar distortion pedal.
- Used circuit modelling methods including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.
- Developed emulation as an audio plugin and guitar pedal using an Arduino Teensy microcontroller.
- Project documentation is available on GitHub.

## Modal Waterbottles

Stanford, CA

Researcher

June 2019 - April 2020

- Worked with a team of researchers to develop acoustical models of water bottles using modal signal processing.
- Developed a real-time water bottle synthesizer using the JUCE framework.
- Project documentation is available on GitHub.

## Complex Nonlinearities

Lead Researcher

Aug. 2019 - June 2020

- Built frameworks for developing complex nonlinear audio signal processors.
- Developed a series of open-source plugins to demonstrate the nonlinear frameworks.
- Published a series of Medium articles explaining nonlinear signal processing techniques for a non-technical audience.
- Project documentation can be found on GitHub.

## Bad Circuit Modelling

Lead Researcher

Nov. 2019 - June 2020

- Researched methods for virtual analog circuit modelling to account for imperfections that exist in real-world circuits.
- Developed simulations and real-time circuit-modelling systems that accurately model the aging and component tolerances of resistors and capacitors.
- Project documentation can be found on GitHub.

## Analog Tape Modelling

Stanford, CA

Lead Researcher

Jan. 2019 - Apr. 2019

- Researched and developed a physical model of an analog, reel-to-reel tape machine.
- Modelled the nonlinear process of analog tape recording using the Jiles-Atherton model for magnetic hysteresis.
- Modelled other effects created by analog tape machines, including flutter, playhead loss effects, and biasing effects.
- Developed an open-source audio plugin implementing the analog tape model as a real-time system. Source code and documentation are available on GitHub.

## notGuitar

Los Angeles, CA

Research Partner

Jan. 2018 - May 2018

- notGuitar is real-time timbral conversion system designed to process a guitar input signal to sound like a saxophone.
- notGuitar was implemented using a Texas Instruments DSK6713 DSP board in May 2018.
- Documentation for the project can be found on GitHub.

# Projects

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## Chowdhury DSP

Founder, Engineer

June 2018 - PRESENT

- Developed open-source audio plugins including:
  - ChowTapeModel: An analog tape emulation based on the DAFx-19 paper “Real-Time Physical Modelling for Analog Tape Machines”.
  - BYOD: A modular system for creating guitar effect signal chains.
  - ChowMatrix: A dynamic matrix of stereo delay effects.
  - ChowKick: A kick drum synthesizer based on creative circuit modelling.
  - ChowCentaur: A digital emulation of the Klon Centaur distortion pedal, using Wave Digital Filters and recurrent neural networks.
  - ChowPhaser: A phaser effect based loosely on the Schulte Compact Phasing ‘A’ circuit.
- Developed effects for guitar pedal platforms including the Aviate Audio Multiverse and Chaos Audio Stratus pedals.
- Contributed audio effects and other DSP code to the open-source Surge Synthesizer project.
- Developed modules for the VCV Rack modular synthesis environment.
- For more information, see the Chowdhury DSP website.

## CLever Audio Plugins (CLAP)

Contributor

Mar. 2022 - PRESENT

- Contributed to the CLAP plugin standard with testing and documentation.
- Proposed new “extensions” to the CLAP standard.
- Contributed to a wrapper to allow developers using the JUCE framework to export CLAP plugins.
- Contributed to a JUCE module allowing JUCE applications to host CLAP plugins.

## Wave Digital Filters

Programmer

Jan. 2020 - PRESENT

- Created a minimal C/C++ Library for modelling circuits with Wave Digital Filters (WDFs).
- Currently includes WDF implementations for basic circuit elements (resistors, capacitors, inductors), basic circuit adaptors (series, parallel), “1-port” nonlinear circuit elements, and “R-Type” adaptors.
- Includes several example WDF models implemented as audio plugins using the JUCE framework, as well as a graphical WDF prototyping tool.
- Developed an auxiliary script for deriving scattering matrices for “R-Type” adaptors.
- Ongoing development and documentation can be found on GitHub.

## NoLava Recording Studios

Audio Engineer

Los Angeles, CA

Aug. 2017 - July 2018

- Recorded, mixed, and mastered for artists of various styles including acoustic, electronic, rock, country, and punk.
- Installed, repaired, and maintained speakers, microphones, keyboards, amplifiers, and other studio equipment.